Influencing, Reading and Extending Spin — in the FFGFT Framework

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.1 Influencing Spin — More than Just Magnetic Fields

In Doc. 173 spin appeared briefly in the context of the Ising machine and the minimal quantum resonator: two permitted orientations of a torsion configuration — spin-up and spin-down. The question of how to rotate this orientation from outside and how to read it out is the bridge between the abstract resonator picture and technical realisation.

The magnetic field is the historically first and most intuitive way — but by far not the only one.

Method 1: Magnetic Field — Zeeman Effect

A static magnetic field **B** splits the two spin states energetically:

$$\Delta E_Z = g^* \mu_B B, \tag{1}$$

where g^* is the effective g-factor of the material and μ_B is the Bohr magneton. For GaAs $g^* \approx -0.44$ — significantly smaller than for the free electron ($g = 2$), because the band structure of the crystal modifies the effective spin-orbit coupling.

A resonant radio-frequency pulse at the Larmor frequency $\nu_L = g^* \mu_B B/h$ rotates the spin in a controlled manner. This is the principle of electron spin resonance (ESR) and — in essence — also of magnetic resonance imaging.

FFGFT language: magnetic field as global torsion field

An external magnetic field establishes a preferred direction in the torsion field. It breaks the symmetry between spin-up and spin-down energetically. The radio-frequency pulse at ν_L is a periodically oscillating torsion perturbation — resonant when its frequency exactly corresponds to the energy difference of the two torsion configurations.

Method 2: Electric Field — EDSR

Electric fields do not couple directly to spin — the electron has no electric dipole moment component in spin space. But indirectly they couple very well, via **spin-orbit coupling**:

The electron moving in the electric field of the crystal lattice sees in its own rest frame an effective magnetic field. This internal field rotates the spin. A rapidly varying external electric field can resonantly drive this mechanism — this is called **EDSR** (Electric Dipole Spin Resonance).

Advantage over magnetic field methods

Electric fields can be generated locally with gate electrodes on a few nanometres and switched in under a nanosecond — magnetic fields cannot. For integrated qubit arrays EDSR is therefore the preferred method for single-qubit rotation.

In FFGFT language: the electric field deforms the resonator geometry locally — and via the spin-orbit coupling this geometry change translates directly into a torsion change of the spin degree of freedom. The coupling is not direct, but geometrically mediated.

Method 3: Circularly Polarised Light

Circularly polarised light (σ^+ or σ^-) carries angular momentum $\pm\hbar$ per photon. Upon resonant excitation of a quantum dot, a σ^+ photon transfers this angular momentum directly to the created exciton — and thus to the electron spin.

This allows **purely optical spin preparation** without any external field: by choice of the photon polarisation one selects the spin state.

Readout is via the polarisation of the fluorescence light: depending on spin state the quantum dot preferentially emits σ^+ or σ^- light. Polarisation measurement $\equiv$ spin measurement.

FFGFT: photon transfers torsion angular momentum

In FFGFT the photon carries its angular momentum as a topological property of the electromagnetic field torsion. Upon absorption this field torsion rotates the electron torsion into the complementary configuration. Spin-up and spin-down are two opposite torsion windings — the circular photon switches between them because it itself carries a directed winding.

Method 4: Exchange Interaction

When two electrons in adjacent quantum dots spatially overlap, their spins couple directly to each other — the **exchange interaction** with strength J:

$$H_J = J(t)\,\mathbf{S}_1 \cdot \mathbf{S}_2. \tag{2}$$

By pulsing the gate field (which controls the overlap and thus $J(t)$) one can execute controlled two-qubit operations.

This is the basis for **spin qubit gates** in semiconductor systems (Loss-DiVincenzo model, 1998). The strength: no external microwave needed — the coupling is purely electrically controllable.

Method 5: Hyperfine Interaction

The electron in the quantum dot interacts with the nuclear spins of the $\sim 10^4$–10^6 surrounding atoms. This coupling is normally a problem: the fluctuating nuclear spin bath creates a random effective magnetic field and destroys the spin polarisation in times of $T_2^* \sim 10$–100 ns.

But used in a controlled manner it is a tool: through dynamic decoupling (fast pulse sequences that average out the nuclear spin fluctuations) T_2 can be extended to microseconds. In some systems (e.g. phosphorus in silicon) the nuclear spin itself is used as a highly stable qubit ($T_1 > 1$ s at low temperatures).

.2 Reading Out Spin — Five Ways

Spin-to-Charge Conversion

The most important method in practice. The potential is chosen such that only spin-down (or only spin-up) is energetically permitted to tunnel out of the quantum dot. A single tunnelling electron creates a measurable current pulse in the adjacent **single-electron transistor** (SET).

No signal: spin-up (blocked). Signal: spin-down (tunnelled). Sensitivity: single electrons, error rate $< 1\,\%$.

Spin-to-charge conversion: state of the art 2025

In Si/SiGe and GaAs systems, spin-to-charge conversion readout schemes achieve accuracies $> 99.5\,\%$ with readout times below $1\,\mu$s — sufficient for error-corrected quantum computers beyond the surface code threshold.

Pauli Blockade

More elegant: no external measuring device needed. The Pauli principle itself is the detector.

Two adjacent quantum dots, one electron each. An attempt is made to load both electrons into the same dot. If the spins are antiparallel: possible (both fit in the ground level). If the spins are parallel: not possible (Pauli exclusion) — the transport is **blocked**.

Measurable as a conductance difference: blockade $\equiv$ same spins.

FFGFT: Pauli blockade as geometric incompatibility

Two identical torsion configurations cannot occupy the same resonator — not because of a postulated prohibition, but because the same torsion winding in a cavity permits no second stable configuration of the same kind. The Pauli principle is in FFGFT a geometric statement about the compatibility of field configurations — not an independent quantisation rule.

Faraday and Kerr Rotation — State-Preserving Readout

Optical readout exploits the fact that spin-up and spin-down couple to different optical transitions. Circularly polarised light is scattered differently depending on spin state — the polarisation of the fluorescence light reveals the spin state. This resonant fluorescence is invasive: the absorbed photon excites the system and changes its state.

Fundamentally different from this is **Faraday / Kerr rotation**: a linearly polarised laser beam passes *by* the quantum dot — it is not absorbed, but rotates its plane of polarisation by a tiny angle $\theta_F \propto S_z$ upon passing. The spin state is readable from the rotation angle, without a single photon being absorbed.

Faraday rotation: reading quantum state without disturbing it

Faraday rotation is a **Quantum Non-Demolition measurement** (QND) — a readout method that does not change the measured state itself.

Physically: the light field couples dispersively to the spin — the interaction is chosen so that it *commutes* with the spin operator S_z:

$$[H_{\text{Faraday}}, S_z] = 0. \tag{3}$$

An observable that commutes with the Hamiltonian is not disturbed by the measurement. The polarisation rotation θ_F is proportional to S_z — the eigenvalue is read off, the eigenstate is preserved.

Time resolution: in the picosecond range. Sensitivity: single spins in quantum dots detectable. The spin state can be measured *repeatedly* — each measurement yields the same result.

FFGFT: QND measurement as proof of stable torsion configuration

The fact that Faraday rotation does not change the spin state is in FFGFT not a special case — it is the direct consequence of the stability of the torsion configuration. Spin-up is a stable topological torsion. A passing photon rotates its plane of polarisation because it passes through the torsion field — but it does not bring the torsion configuration out of its stable state, because the coupling is so weak and symmetric that no energy transfer into the torsion mode occurs.
This is the same logic as with a resonator: a weak signal that does not hit the resonance frequency passes through the resonator without exciting it. The photon of the Faraday measurement intentionally does *not* hit the resonance frequency of the transition — it reads off the topological invariant without disturbing it.
The decisive conclusion: When a measurement repeatedly yields the same state and in doing so does not change the state, then the state had that value *before the measurement*. This is the empirical proof that stable torsion configurations really and definitely exist — independently of whether we measure them or not. The QND measurement proves determinism.

.3 Beyond Two States — and How to Overcome the Limit

A spin qubit has two states: $|\uparrow\rangle$ and $|\downarrow\rangle$. This is the absolute minimum of a quantum resonator (Doc. 173, section on the minimal resonator).

But a quantum dot is not a two-state system by nature — it has arbitrarily many energy levels. The spin is only the *simplest* feature.

There are at least five strategies for extracting more than two distinguishable states from a quantum dot for quantum information.

Strategy 1: Qutrit and Qudit — More Energy Levels

Instead of $|0\rangle$ and $|1\rangle$ one uses $|0\rangle$, $|1\rangle$, $|2\rangle$ — a **qutrit** (three states) — or generally $|0\rangle$ to $|d-1\rangle$: a **qudit** of dimension d.

In the quantum dot this corresponds to the first d energy levels. The basic formula:

$$E_n = \frac{n^2\pi^2\hbar^2}{2m_{\mathrm{eff}}R^2}, \quad n = 1, 2, \ldots, d. \tag{4}$$

For $d = 3$ (qutrit) one needs three distinguishable levels with $E_3 - E_2 \gg k_B T$ — otherwise the thermal population blurs the third level.

Condition for a stable qudit of dimension d

All d level spacings must exceed the thermal energy:

$$E_{n+1} - E_n = (2n+1)\,E_1 \gg k_B T \quad \text{for all } n = 1, \ldots, d-1.$$

The most critical spacing is $E_2 - E_1 = 3E_1$ (the smallest). Qudits of high dimension require small quantum dots or low temperatures — or both.

The information density of a qudit: $\log_2(d)$ bits per object. A qutrit encodes $\log_2(3) \approx 1.585$ bits — 58 % more than a qubit per object. A qu*five*-it encodes $\log_2(5) \approx 2.32$ bits.

Qudit efficiency at ξ level $N \approx 8$

A quantum dot with $R = 3$ nm at 4 K has at least 5 thermally stable levels (all spacings $\gg k_B T(4\,\mathrm{K}) = 0.34$ meV). This corresponds to a qu*five*-it with $\log_2(5) \approx 2.32$ bits of information content — in a resonator of 22 nm diameter.

Strategy 2: Orbital Qubit — Geometry Instead of Energy

A quantum dot not only has different energy levels — it also has different **orbitals** (spatial wave patterns with the same energy, but different geometry): s-, p-, d-like states as in the atom.

In a non-spherical quantum dot the p-orbitals are energetically split. One can use the two lowest p-states (p_x and p_y) as a qubit — an **orbital qubit** — *without* looking at the spin.

The advantage: orbital states couple strongly to electric fields, weakly to magnetic noise. This makes them potentially more coherent in magnetically noisy environments.

Strategy 3: Exciton Number Qudit

From Doc. 173: a quantum dot can carry $0, 1, 2, 3, \ldots$ excitons simultaneously — discrete storage levels, sharply distinguishable.

This exciton number itself is a qudit:

$$|0\,\mathrm{Ex}\rangle, \quad |1\,\mathrm{Ex}\rangle, \quad |2\,\mathrm{Ex}\rangle, \quad |3\,\mathrm{Ex}\rangle, \ \ldots$$

Readout is via the emission wavelength: a biexciton ($|2\,\mathrm{Ex}\rangle$) emits at a different frequency than an exciton ($|1\,\mathrm{Ex}\rangle$) — spectrally clearly distinguishable.

FFGFT: exciton number as torsion pair number

Each exciton is a bound torsion-anti-torsion pair in the resonator. The exciton number is the number of such pairs — a topological invariant of the field configuration. Adding an exciton means adding a new torsion-anti-torsion pair to the resonator. The discreteness of the exciton number follows directly from the topological nature of the pairs: you cannot add half a torsion pair.

Strategy 4: Spin-Orbit Qudit — Spin and Orbital Combined

The most powerful combination: spin *and* orbital degree of freedom are used simultaneously.

A quantum dot with spin (↑/↓) and two orbital states (p_x/p_y) has four combined states: $|p_x, \uparrow\rangle$, $|p_x, \downarrow\rangle$, $|p_y, \uparrow\rangle$, $|p_y, \downarrow\rangle$ — a **ququart** (qudit of dimension 4, equivalent to 2 entangled qubits in a single object).

The spin-orbit coupling connects the two degrees of freedom and allows navigating between the combined states through electric fields.

Strategy	Degree of freedom	Dimension d	Bits per QD
Spin qubit	Spin	2	1.00
Orbital qubit	Orbital	2–4	1.00–2.00
Qudit	Energy level	3–6	1.58–2.58
Exciton qudit	Exciton number	3–5	1.58–2.32
Spin-orbital	Spin + orbital	4–8	2.00–3.00

Strategy 5: Continuous Variables — the Quantum Mechanical Oscillator

A radical alternative to the discrete qudit concept: instead of discrete states one uses **continuous variables** (CV).

A harmonic oscillator has infinitely many energy levels $|n\rangle$, $n = 0, 1, 2, \ldots$ The information is not stored in individual levels, but in the **waveform** of the state — for example as a coherent state $|\alpha\rangle$ with complex amplitude α.

Quantum dots are *not* harmonic oscillators (their level spacings are not equidistant: $E_n \sim n^2$) — but **optical cavities** and **superconducting resonators** (Circuit-QED) are approximately.

Discrete or continuous — a choice of description level

In FFGFT both descriptions are facets of the same physics. Discrete states arise from the finite geometry of the resonator — the resonator is finite, so its modes are countable. An infinite number of modes (ideal harmonic oscillator) is the limiting case of an infinitely deep potential with uniform walls — an idealisation. Real systems are always discrete. Continuous variables are a useful approximation when the number of modes is so large that they can be treated as continuous — analogous to classical physics as the limit of many quantum states.

.4 Superposition — What Lies Between the States

So far: discrete states $|0\rangle$, $|1\rangle$, ...What is special about quantum systems is not that they have discrete states — classical coins do too. What is special is **superposition**:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \tag{5}$$

This is not ignorance about the state — it is a genuine intermediate state.

On the Bloch sphere (mapping of qubit states onto a sphere surface) every point corresponds to a possible quantum state — infinitely many continuous points on the sphere, not just the two poles.

FFGFT: superposition as intermediate torsion

Spin-up and spin-down are the two stable torsion extremes of the resonator. A superposition state $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$ is a **metastable intermediate torsion** — not stable in the ground state, but stable on the timescale of the coherence time T_2.
The phase $\phi = \arg(\alpha/\beta)$ is the rotation position of this intermediate torsion. Decoherence is the process in which the environment pushes this intermediate torsion into one of the two stable extreme configurations.

For a qudit of dimension d the general state lies on a $(d-1)$-dimensional complex sphere — the $\mathbb{CP}^{d-1}$. With growing d the state space grows exponentially: n qudits of dimension d span a space of dimension d^n.

.5 Decoherence — the Enemy of Superposition

Superposition is fragile. The environment measures constantly alongside — every interaction with a phonon, a nuclear spin, a photon is an (unwanted) measurement that collapses the state.

Two timescales:

T_1 **(relaxation time)**: time until the system has fallen from the excited state into the ground state. Energy relaxation — the state $|1\rangle$ becomes $|0\rangle$.

T_2 **(coherence time)**: time until the phase ϕ of the superposition is destroyed. Always $T_2 \leq 2T_1$ (phase relaxation is at least as fast as energy relaxation, usually faster).

System	T_1	T_2	**Main mechanism**
Electron spin in GaAs	~ 1 ms	~ 10 ns	Nuclear spin bath
Electron spin in Si/SiGe	> 1 s	~ 10 ms	^{28}Si (nuclear spin-free)
Electron spin in P:Si	> 1 s	> 100 ms	Isotopic purification
Exciton in CdSe QD	~ 1 ns	~ 10 ps	Phonon scattering
Superconductor transmon	~ 100 µs	~ 100 µs	Dielectric losses

Si/SiGe: FFGFT significance of $T_2 \sim 10$ ms

Silicon with enriched ^{28}Si (nuclear spin-free) eliminates the nuclear spin bath almost completely. In FFGFT language: the resonator is freed from a source of random torsion perturbations (nuclear spins). What remains is the intrinsic geometry of the resonator — and thus the fundamental limit $T_2 \leq 2T_1$. The long coherence time in ^{28}Si is a direct demonstration that Condition 1 (geometric perfection) is the decisive one.

.6 Open Connections to FFGFT

The following points have not yet been worked out — they mark places where the resonator picture of FFGFT could make concrete predictions:

Open questions for further chapters

1. Qudit level structure and K_{frak}:
The fractal correction factor K_{frak} (Doc. 173) shifts all energy levels by the same relative amount. For a qudit of dimension d this means $d-1$ shifted transitions — a structured spectral signature that deviates from the ideal parabolic. Is this deviation measurable with modern multiphoton spectroscopy?

2. Spin-orbit coupling as torsion-torsion coupling:
The spin-orbit coupling connects two geometric degrees of freedom (spin = torsion direction, orbital = torsion pattern). In FFGFT this coupling should be material-dependent — exactly via the effective g-factor g^* and the band structure torsion of the crystal. Can g^* be derived from ξ and the crystal geometry?

3. Decoherence timescales and ξ hierarchy — numerical analysis:
Measured coherence times at the ξ level $N \approx 8$:

System	Timescale	Ratio to exciton
Exciton (CdSe QD)	$T_2 \sim 10$ ps	1 (reference)
Spin qubit Si/SiGe	$T_2 \sim 10$ ms	10^9 (= 9 orders of magnitude)
Nuclear spin P:Si	$T_1 > 1$ s	10^{11} (= 11 orders of magnitude)

Numerical comparison with the two ξ parameters:

Parameter	$\log_{10}(1/\xi)$	2 levels =
$\xi = 1.333 \times 10^{-4}$ (scale hierarchy)	3.875 orders of magnitude	7.75
$\xi_{\text{Higgs}} = 1.038 \times 10^{-5}$ (coupling parameter)	4.984 orders of magnitude	**9.97**

Finding: With ξ_{Higgs} exactly **2 hierarchy levels** span exactly 9.97 orders of magnitude — which lies between the two measured ratios (9 and 11 orders of magnitude) and agrees with the Si/SiGe value to $< 5\,\%$.
Physical interpretation: The exciton resonator at ξ level $N \approx 8$ couples *directly* to its environment (phonons at the same level): decoherence rate $\Gamma_{\text{exciton}} \propto \xi^0 \cdot \omega$. The spin qubit at the same length scale couples to a *remote* bath (nuclear spins at level $N \approx 7$): $\Gamma_{\text{spin}} \propto \xi^2_{\text{Higgs}} \cdot \omega$ — two levels weaker, so $T_2 \propto 1/\xi^2_{\text{Higgs}} \approx 10^{10}$ times longer.
Conclusion and explanation: The agreement of 2 hierarchy levels with ξ_{Higgs} and the measured T_2 ratio is not coincidence — it follows from the **nature of the two FFGFT spaces**:
ξ_{Higgs} acts in the *number space / algebraic field space* and describes couplings between quantum states. Decoherence is a process in state space — a transition between quantum states caused by coupling to the environment. Therefore ξ_{Higgs} is the correct parameter for decoherence rates.
ξ acts in *geometric 3D space* and describes length hierarchies. Coherence times are not geometric lengths — they are algebraic quantities in state space. It would be physically wrong to use ξ for decoherence rates.
The two ξ parameters are **not a contradiction**, but complementary descriptions of different aspects of FFGFT: geometry of space (ξ) and algebra of state space (ξ_{Higgs}).

4. Entanglement as torsion topology:
Two entangled qubits are in a state that cannot be written as a product of two individual states — a common torsion configuration over spatially separated resonators. What does this mean for the local topology of the FFGFT field? Is there a natural description of Bell states as toroidal field configurations?

.7 Summary: From Spin to Qudit

Conclusion

Spin-up and spin-down are the minimum — two torsion orientations in one resonator.
Manipulation: magnetic field (global), electric field via spin-orbit coupling (local, fast), circular light (angular momentum transferring), exchange interaction (two-QD gate).
Readout: spin-to-charge conversion (practical, precise), Pauli blockade (geometrically self-reading), fluorescence polarisation (invasive, time-resolved), **Faraday/Kerr rotation (QND — state-preserving, repeatable)**.
The QND measurement is the physically most important point: it proves that the spin state had a defined value before the measurement — the torsion configuration is real and stable, independent of observation. This is the empirical finding for determinism.
Beyond two states: energy level qudits, orbital qubits, exciton qudits, spin-orbital combination — all in the same tiny resonator, all at the ξ level $N \approx 8$.
Superposition lies between the states — not ignorance, but a genuine intermediate state (intermediate torsion). Decoherence is its natural enemy: the environment collapses the torsion into one of the stable extreme configurations.

Geometric perfection (Condition 1) is the universal foundation — for every one of these effects. Without a precise resonator: no qudit, no qubit, no spin gate. The FFGFT interpretation: all these phenomena are different expressions of stable torsion configurations in geometrically precise cavities at the scale $N \approx 8$.

Bibliography

[1] Loss, D. & DiVincenzo, D. P. *Quantum computation with quantum dots.* Phys. Rev. A **57**, 120–126 (1998).

[2] Koppens, F. H. L. et al. *Driven coherent oscillations of a single electron spin in a quantum dot.* Nature **442**, 766–771 (2006).

[3] Nowack, K. C. et al. *Coherent control of a single electron spin with electric fields.* Science **318**, 1430–1433 (2007).

[4] Corna, A. et al. *Electrically driven electron spin resonance mediated by spin–valley–orbit coupling in a silicon quantum dot.* npj Quantum Inf. **4**, 6 (2018).

[5] Atatüre, M. et al. *Quantum-dot spin-state preparation with near-unity fidelity.* Science **312**, 551–553 (2006).

[6] Press, D. et al. *Complete quantum control of a single quantum dot spin using ultrafast optical pulses.* Nature **456**, 218–221 (2008).

[7] Petta, J. R. et al. *Coherent manipulation of coupled electron spins in semiconductor quantum dots.* Science **309**, 2180–2184 (2005).

[8] Elzerman, J. M. et al. *Single-shot read-out of an individual electron spin in a quantum dot.* Nature **430**, 431–435 (2004).

[9] Ono, K. et al. *Current rectification by Pauli exclusion in a weakly coupled double quantum dot system.* Science **297**, 1313–1317 (2002).

[10] Atatüre, M. et al. *Observation of Faraday rotation from a single confined spin.* Nature Phys. **3**, 101–105 (2007).

[11] Berezovsky, J. et al. *Picosecond coherent optical manipulation of a single electron spin in a quantum dot.* Science **320**, 349–352 (2008).

[12] Grangier, P., Levenson, J. A. & Poizat, J. P. *Quantum non-demolition measurements in optics.* Nature **396**, 537–542 (1998).

[13] Delbecq, M. R. et al. *Quantum non-demolition measurement of an electron spin qubit.* Nature Nanotechnol. **14**, 446–450 (2019).

[14] Xue, X. et al. *Quantum non-demolition readout of an electron spin in silicon.* Nature Commun. **11**, 1178 (2020).

[15] Veldhorst, M. et al. *An addressable quantum dot qubit with fault-tolerant control-fidelity.* Nature Nanotechnol. **9**, 981–985 (2014).